

CHAPTER 10

INTRODUCTION TO VIBRATION ISOLATION MECHANISMS

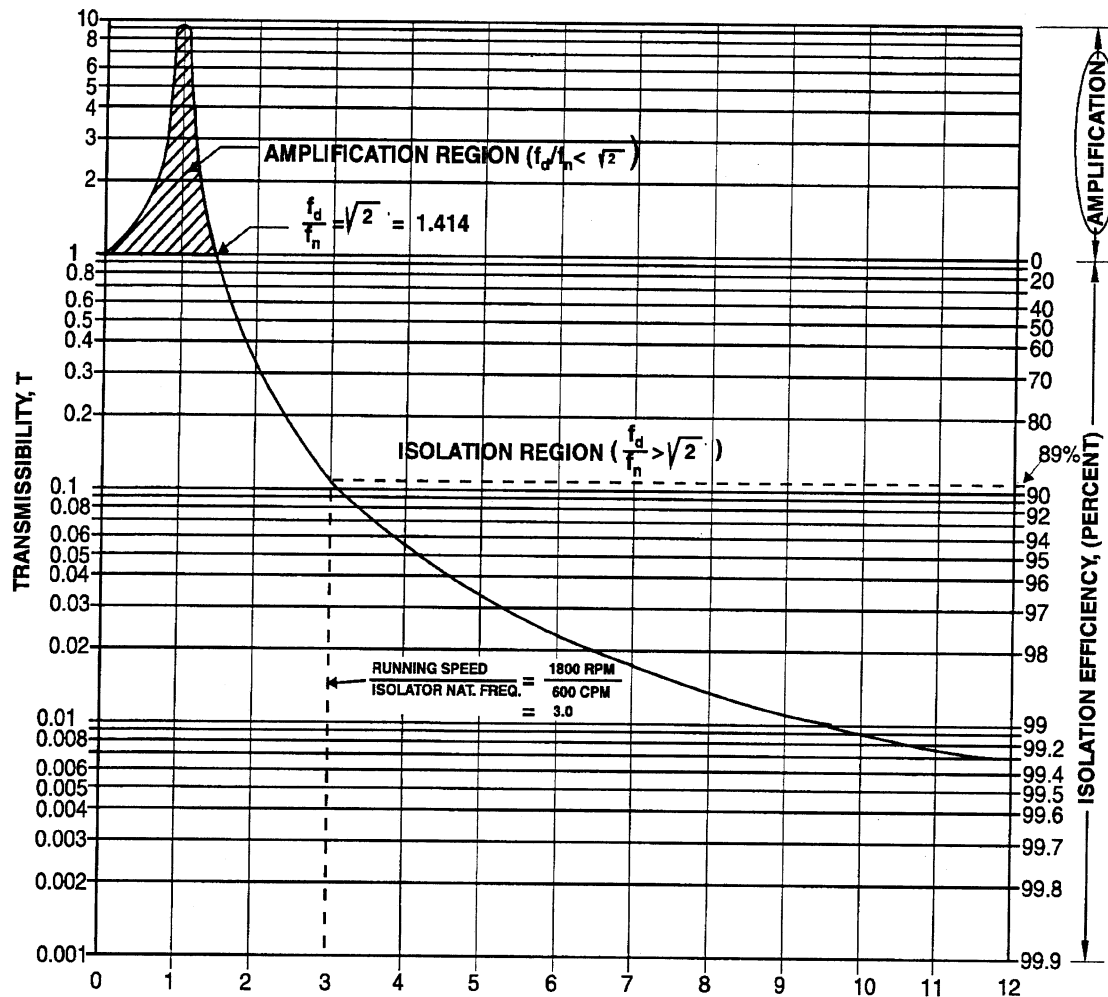
Often, vibration analysts and those not associated with vibration technology will erroneously use the terms "vibration isolation" and "vibration damping" interchangeably. In reality, these are two distinctly different mechanisms having two different purposes and results. First, let us take a look at the definition of vibration isolation and point out how it differs from vibration damping:

- **Vibration Isolation** - The major, overriding purpose of vibration isolation is to prevent the unwanted transmission of vibration from one structure or machine to another by "mismatching" the vibration forcing frequencies of the machine or structure from the natural frequency of the isolator mechanism (see Figure 1). One of the key points is that placing isolators between a machine and supporting floor or other structure will most often not reduce the vibration within the machine (in fact, it will increase vibration levels in most machines when they are placed on isolators, typically by about 20% to 40%). Instead, it will simply reduce the transmission of vibration from this machine to surrounding structures; or, on the other hand, will reduce the incoming vibration from other sources into the machine or structure which is to be isolated. It should be pointed out that even though vibration often increases on a machine when it is placed on isolators, no more damage is being done since no additional forces have been applied to the machine - it now simply has less dynamic resistance under the same loads to which it was formerly being subjected (recall that vibration amplitude is proportional to "dynamic load divided by dynamic resistance").

Damping is still somewhat related to isolation due to the fact that the higher the damping, the lower will be the vibration response when the forcing frequencies within the machine to be isolated happen to coincide with the isolator natural frequency, either as the machine comes up to speed or coasts to a stop. However, as the damping of an isolator increases, its effectiveness will diminish (see Figure 2). Damping itself is "the dissipation of vibratory energy with motion or with time, converting this energy from mechanical motions to other types of energy, chiefly heat". Damping treatments are usually only effective when the structure on which they are placed is vibrating at, or very near, resonance.

- **Why are Isolators Needed?** - One of our client plants in the northeast USA has a total of ten (10) large Roots blowers (lobe-type). Normally, he only needs to operate five of the ten machines. In so doing, he found that he began having problems, not necessarily with the machines which were running, but instead, with the machines which were sitting idle most of the time. Upon investigation, he found that the rolling element bearings in the idle machines were being subjected to "false brinelling", which leaves its "fingerprints" behind as a series of indentations particularly in the load zone of the outer race, with the spacing of the indentations equal to the spacing between each of the rolling elements. If this false brinelling is allowed to continue, these indentations can become quite pronounced. Therefore, when it was necessary to operate these machines, the bearings themselves already could have significant damage on their outer races, which could easily transfer into the rolling elements, and from the rolling elements into the inner race of the bearing bringing about very short-lived life of these bearings. To prevent this from reoccurring required him to install vibration isolation systems on each of the ten Roots blowers in this case. Since doing so, this has virtually eliminated the false brinelling problems and premature bearing failures these machines were suffering before.

FIGURE 1
THEORETICAL VIBRATION ISOLATION EFFICIENCY &
TRANSMISSIBILITY CURVE (ASSUMED DAMPING FACTOR $\zeta = .05$)
(Ref. 1)



$$\text{RATIO} = \frac{\text{FORCING FREQUENCY}}{\text{NATURAL FREQUENCY}} = \frac{f_d}{f_n}$$

Driving Frequency $\swarrow$
 Isolator Natural Frequency $\nwarrow$

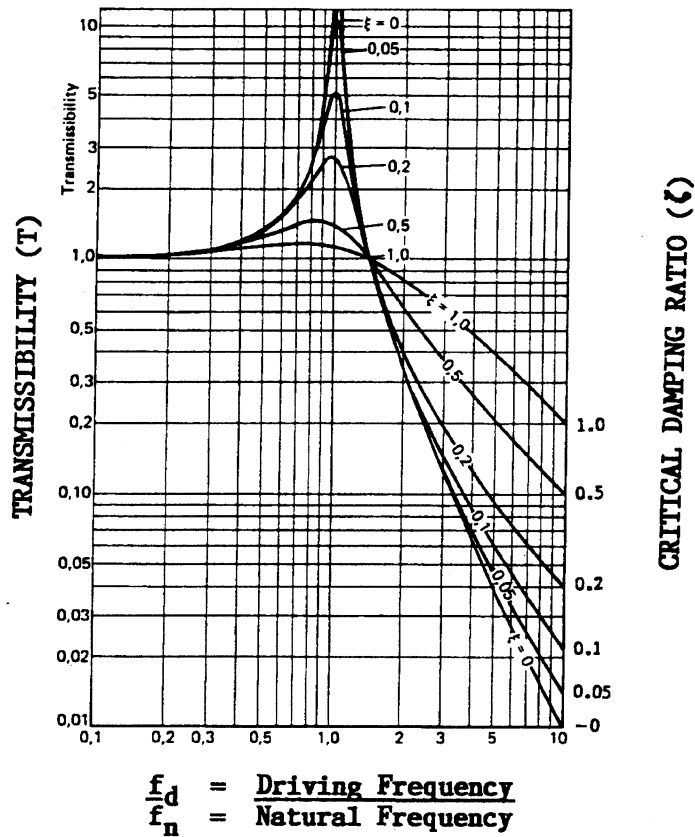
EXAMPLE: IF OPERATING SPEED = 1800 (30 Hz) FOR A DIRECT - COUPLED FAN AND ISOLATOR NAT. FREQ. = 600 CPM (10 Hz)

FIND: THEORETICAL ISOLATION
 FROM GRAPH ABOVE, ISOLATION = 89% (FOR A 3:1 RATIO)

IF AN ISOLATOR WITH A NATURAL FREQ. OF 300 CPM (5 Hz) COULD BE INSTALLED:

$$\frac{1800 \text{ RPM}}{300 \text{ CPM}} = 6:1; \text{ THUS, ISOLATION} = \underline{97.2\%} \text{ (FOR A 6:1 RATIO)}$$

FIGURE 2
THEORETICAL TRANSMISSIBILITY CURVE FOR A RANGE OF DAMPING
FACTORS (ζ) IN A SINGLE DEGREE-OF-FREEDOM SYSTEM
(Ref. 2)



$$\text{TRANSMISSIBILITY} = T = \frac{1 + \left[(2\zeta) \left(\frac{f_d}{f_n} \right) \right]^2}{\sqrt{\left[1 - \left(\frac{f_d}{f_n} \right)^2 \right]^2 + \left[(2\zeta) \left(\frac{f_d}{f_n} \right) \right]^2}}$$

where:

T = Transmissibility
 f_d = Driving or Forcing Frequency
 f_n = Natural Frequency
 ζ = c/c_c = Critical Damping Ratio

In other cases, it might be necessary to install a highly precision measurement instrument such as an optical comparator or an electron microscope in a laboratory where the floor is subjected to noticeable vibration. This vibration might be generated by process machinery located outside the lab such as reciprocating air compressors, punch presses, air handling systems, etc. In several cases, the author has been involved when railroad trains operating up to one-half mile away induced sufficient vibration into a precision lab to totally interrupt any measurement processes with his instrument while the train was passing. In these cases, precision measurements had to be made to evaluate the vibration severity; and then, sophisticated vibration isolation mechanisms had to be put in place to virtually eliminate this undesirable transmission of vibration from such background sources. An illustration of one of these isolation mechanisms will be given later in this section using Figures 9 and 10 showing how a fairly simple isolation system was used to prevent fairly substantial vibration originating at a two-stage reciprocating compressor from transmitting into a load bearing wall located less than two feet away from the compressor; and from transmitting this vibration into an electron microscope located directly above this compressor installation on the second floor.

- **How Does Vibration Isolation Work?** - Figure 1 helps explain how vibration isolation works (this figure assumes a damping factor, ζ , equal to .05). In Figure 1, the left vertical axis graphs the vibration "Transmissibility" (T) while the right-hand vertical side graphs "Isolation Efficiency Percentage". The horizontal axis pertains the ratio of forcing frequency (f_d) divided by the natural frequency (f_n). In this case, the natural frequency will actually be the natural frequency of the entire isolated system, but will be almost totally influenced and controlled by the natural frequency of the isolation mechanism itself. In the case of forcing frequency, sometimes called driving frequency (f_d), this frequency can be due not only to operating speed, but also to 2X RPM, 3X RPM, blade pass frequency (#blades X RPM), gear mesh frequency (#gear teeth X RPM), etc. The key point here will be to find the lowest forcing frequency of considerable amplitude and to design the entire isolation system around this lowest frequency of significant amplitude. Whereas in most machines this frequency will normally be at operating speed, please be careful if working with a belt-driven system (particularly a multi-belt driven machine) since the "belt RPM" will always be lower than the RPM of either the driver or driven machine components. When the belts begin to wear, or if they are either severely mismatched or operating at or near a belt resonance, they themselves can be the source of considerable vibration which must be taken into account when specifying a vibration isolator. The key point here is that assuming a single degree of freedom system such as that graphed in Figure 1 (one mass and one spring), if the isolator chosen effectively isolates the lowest frequency of significant amplitude, it will do an even more effective job on isolating higher and higher frequencies. For example, consider a 1780 RPM motor driving a 1000 RPM fan using a six V-belts which themselves turn at approximately 300 RPM. Again, if the belts encounter problems, this frequency which would have to be isolated. If this machine happened to operate near precision instruments, the lowest frequency of significant amplitude very likely might not be the 1000 RPM fan speed, but the belt speed of approximately 300 RPM instead. It should be pointed out that the lower the frequency which must be isolated, the more difficult will be the isolator requirements.

- **What is a Good Rule of Thumb for Specifying Proper Vibration Isolators?** - First, please return to Figure 1. Note on the upper left-hand side of the figure the region denoted as the "Amplification Region". Unfortunately, in plants throughout much of the world, vibration "amplifiers" have been installed rather than vibration isolators. That is, both the machine and the surrounding area would be much better off without any so-called "isolators" since they are in reality "amplifiers". That is, if the forcing frequency of either the machine to be isolated, or of the vibrations transmitting through the floor or surrounding structure into this machine happen to line up with the isolator natural frequency ($f_d/f_n = 1$), the vibration can easily be 10 or more times higher than it would have been were no "isolators" installed whatsoever. In fact, no isolation takes place until the lowest forcing frequency (f_d) is at least 1.414X the isolator natural frequency (f_n). Figure 1 helps demonstrate a good rule of thumb for general rotating machinery. In this case, if an

isolator can be found where the forcing frequency is at least 3X higher than the isolator natural frequency, fairly significant isolation on the order of approximately 89% will be achieved. In other cases, where even more isolation is required, this ratio might have to be increased to as much as 7 or more to achieve about 98% isolation which is required in the case of many precision instruments. Looking at the example given in Figure 1, note here that this was a direct-coupled fan which was transmitting considerable vibration into the surrounding structure. In this case, an isolator with a natural frequency of 600 CPM (10 Hz) was used under the motor/fan system operating at 1800 RPM which resulted in 89% isolation.

- **How Does the Amount of Isolator Damping Affect Isolator Performance?** - Some isolators can be outfitted or combined with damping materials which helps reduce the vibration response when the machine accelerates to operating speed through particularly the isolator natural frequency. In fact, some steel spring systems must be outfitted with snubbers to prevent the motion from becoming excessive if the machine takes some time to pass through this resonance. The problem with adding damping is that the higher the damping, the less will be the resultant isolation. For example, at a ratio of forcing frequency over natural frequency of 10.0, an isolator with a damping ratio of .02 will theoretically have an isolation of 99.0%; while an isolator with a much higher damping ratio of .25 will have an isolation percentage of about 94% (this may not appear to be much of a difference, but if on a precision gage, it may make the difference). On the other hand, if the frequency ratio were only 3.0, this would theoretically drop the isolation from approximately 90% down to only 75% (meaning that 25% of the vibration would pass through with the higher damped isolator compared with only 10% in the case of the lower damped isolator).

- **What Are Some Typical Types of Isolators and How Does Their Performance Compare?** Table I compares some of the more common isolator types, along with their normal range of natural frequencies. Likewise, Table I shows what range of theoretical forcing frequencies each isolator type will isolate, both for approximately 90% as well as for 96% isolation. That is, if one had a 1780 RPM, direct-coupled fan, he would definitely not want to choose either cork or ribbed elastomeric pads for isolators as shown in Table I. Instead, he should choose either steel springs, elastomeric isolators or fiberglass pads. On the other hand, if he were trying to isolate a lower speed, belt-driven fan of approximately 450 RPM, he would likely be forced to use steel springs or pneumatic mounts as shown in the table. Figures 3 through 7 show some of the typical isolator types used today ranging from the rather inexpensive isolators shown in Figures 3 and 4 up to the costly pneumatic spring systems which are supplied with a continuous air supply as shown in Figure 7. The less expensive isolators such as the steel spring types shown in Figure 5 can be used on machines operating at speeds of 400 RPM and above. However, if the speed drops down to 100 or 200 RPM (or if high percentage isolation is required for a precision instrument) the more expensive pneumatic spring isolators which are provided with continuous air supply will likely be required. These particular units have natural frequencies themselves as low as 0.5 Hz (30 CPM). Most of these mechanisms are self-leveling and self-monitoring to keep each of the air springs supplied with just the right amount of air for both leveling and optimum isolating performance. Using these devices, it is possible to make precision measurements with the instruments, even when the plant might be located near a railroad track, or if there are production machines nearby causing much vibration transmission through the floor such as reciprocating machines, punch presses, etc.

TABLE I.
APPROXIMATE NATURAL FREQUENCY RANGES & PERFORMANCE DATA
FOR VARIOUS ISOLATOR TYPES AND MATERIALS

ISOLATOR TYPE	NATURAL FREQ. RANGES (Hz)	FORCING FREQUENCIES THEY WILL ISOLATE			
		≈ 90% ISOLATION		≈ 96% ISOLATION	
		Hz	CPM	Hz	CPM
Cork	20 - 40	66 - 132	3960 - 7920	100 - 200	6000 - 12000
Ribbed Elastomeric Pads	15 - 35	50 - 115	3000 - 6900	75 - 175	4500 - 10500
Fiberglass Pads	5 - 15	17 - 50	1020 - 3000	25 - 75	1500 - 10500
Elastomeric Isolators	6 - 12	20 - 40	1200 - 2400	30 - 60	1800 - 3600
Steel Springs	2 - 8	7 - 26	420 - 1560	10 - 40	600 - 2400
Pneumatic Mounts (Non-Cont. Air)	2 - 4	7 - 13	420 - 780	10 - 20	600 - 1200
Pneumatic Mounts (Continuous Air)	0.5 - 1.5	1.7 - 5	102 - 300	2.5 - 7.5	150 - 450

NOTE: Natural Frequency Ranges and forcing frequencies for which they will provide 90% and 96% isolation listed above for each Isolator Type are approximate ranges which depend on exact design configuration and load distribution. Consult isolator manufacturer for more precise natural frequency and performance data.

FIGURE 3
RIBBED ELASTOMERIC PAD ISOLATOR
 (Ref. 3)

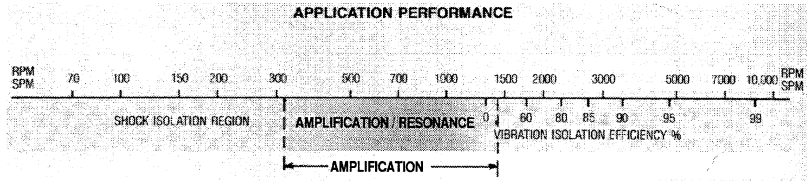
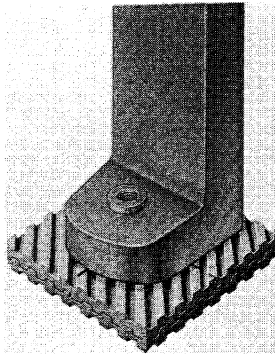
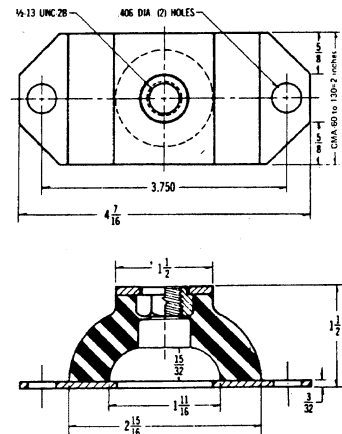
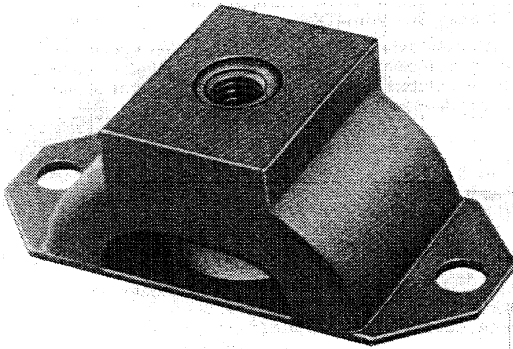


FIGURE 4
ELASTOMERIC ISOLATOR
 (Ref. 3)



Installation: The top plate includes a captivated nut tapped 1/2-13 UNC-2B for ease of attaching equipment. Through holes on each side flange accommodates 5/8" tie-down bolts for secure attachment to support structure. (Bolts not supplied.)

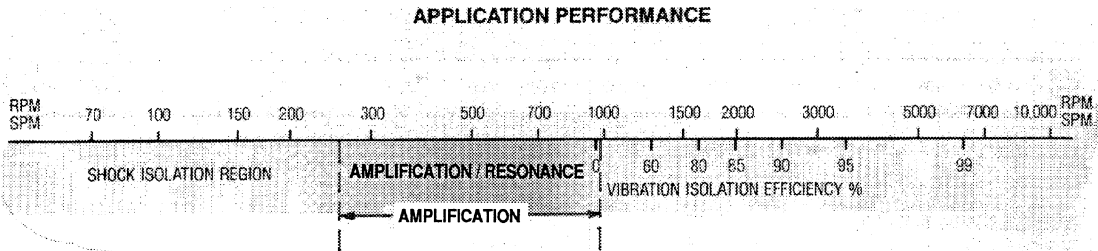


FIGURE 5
STEEL SPRING ISOLATOR WITH LEVELING BOLT
AND ADJUSTABLE SNUBBING
(Ref. 4)

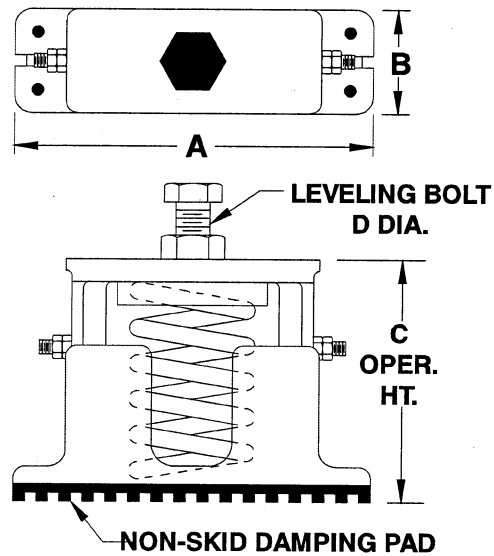


FIGURE 6
PNEUMATIC MOUNTS (NONCONTINUOUS AIR SUPPLY)
(Ref. 3)

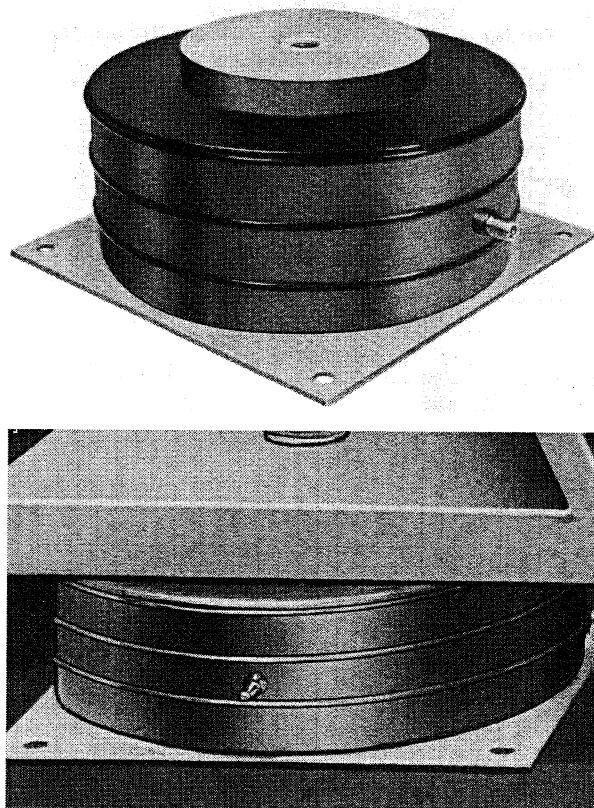


FIGURE 7
PNEUMATIC SPRING WITH CONTINUOUS AIR SUPPLY FOR
VERY LOW FREQUENCY ISOLATION & FOR PRECISION
MEASUREMENT INSTRUMENTS (Ref. 5)

The use of electron microscopes in research, manufacturing, and quality assurance may require that equipment be placed in upper floor areas close to mechanical equipment rooms, elevators, public areas, or near airports, highways, railroads, or industrial areas, where high vibration levels exist.

Because of the extremely high resolution capabilities of electron microscopes, high maintenance costs or blurred images may result, and high magnification may not be possible due to the effect of vibration.

Electron Microscope Isolation
Systems consist of a 2", 3", or 4" thick structural plate base, flame cut to the contour of the microscope base, supported 1" off the floor by three or four air springs. Outboard brackets are used to reduce mounting height at each isolator location. Auxiliary air reservoirs are used to increase damping and to lower the natural frequency. Air regulator level control is most generally used, although fluidic level control is available as an option.

The Isolation System can be operated with 100 PSI building air or with a nitrogen bottle placed near the microscope.

Air Supported Isolation
Systems are available for most standard Electron microscopes, including AMR, ETEC, HITACHI, JEOL, PHILIPS, and RCA, and can be designed and supplied for microscopes of other manufacturers.

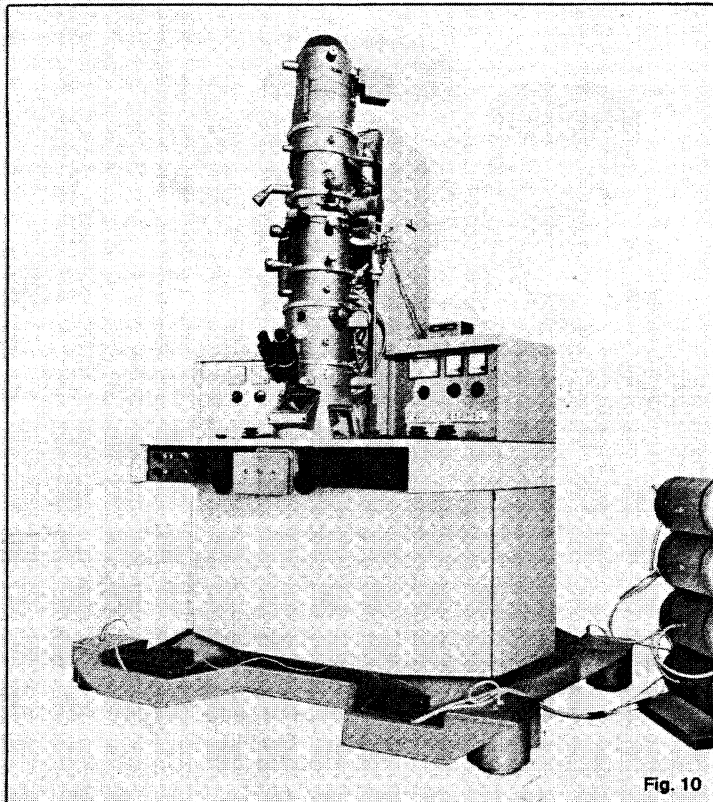


Fig. 10

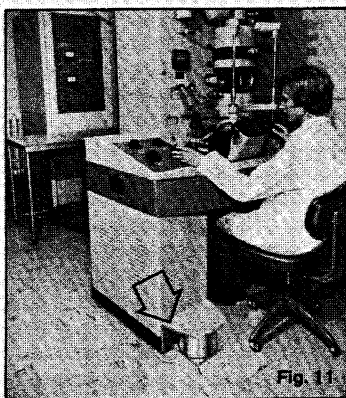


Fig. 11

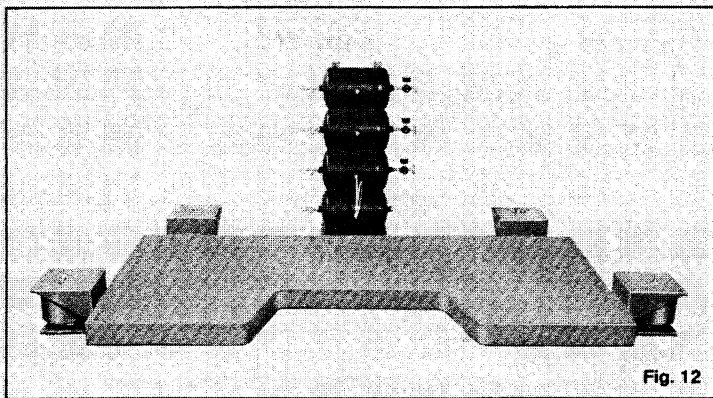


Fig. 12

● **Real-World Case History - "Provision of an Effective Isolation System to Prevent Transmission of Vibration into an Electron Microscope from a 2-Stage Reciprocating Air Compressor to be Installed on the First Floor Directly Beneath the Microscope Lab"**

In practical situations, it is not always necessary to expend great funds when needing significant isolation. It only requires some common sense and practical experience. For example, please refer to the isolation system provided for the two-stage reciprocating air compressor pictured in Figures 8 and 9. In this case, this was a used compressor purchased by the client who allowed the compressor to sit out in the weather for over one year before its installation. Then, he decided to locate it on the first floor of a building, only 21" away from a load bearing wall as shown in Figure 9. Right above the compressor on the second floor was an electron microscope. Therefore, several measures were taken to minimize the vibration transmission into the floor and surrounding structure. First, an entire floating isolation frame was fabricated from channel and I-beams as shown in Figure 9. Then, 15" of concrete was poured into the isolator frame creating an inertia block some 72" wide and 126" long. This entire inertia block and isolation frame was then supported by steel spring isolators with leveling bolts and adjustable snubbing like that shown in Figure 5. Also, a number of calculations were made to ensure that the system center of gravity ("SYSTEM CG" shown in Figure 8) would be positioned to be in line with the center of flexure of each of the spring isolators (note "CF" which shows the spring center of flexure in Figure 8). By aligning the system CG and spring center of flexure (CF), this minimized rocking when the compressor was started up and brought up to its 600 RPM operating speed. During this time, it of course passed through the isolator natural frequency which was calculated to be 2.27 Hz (136 CPM). Comparing this with the 600 RPM operating speed, this worked out to a ratio of about 4.4. Referring back to Figure 1, this resulted in an isolation efficiency of approximately 95%. In addition, before installing the compressor isolation frame, the original floor and the top layers of soil were removed. The upper soil layers in the pit were then compacted to a compaction rate of about 90%. Then, layers of polystyrene were inserted before repouring a new concrete floor, but this time with a 1" cork expansion joint installed all the way around the concrete pad as shown in Figure 9. Finally, a layer of .25" thick neoprene vibration damping pads were placed between the concrete floor and each of the isolator spring support stands to help dampen vibrations which did transmit through the isolators. Also, the client was told to maintain at least a 2" clearance between the floor and the floating inertia block to prevent trash and other foreign materials from building up over time, thereby "short circuiting" the isolation mechanism. Finally, flexible connections similar to those shown in Figures 10 and 11 were placed in all piping connected to the compressor system, again to prevent short circuiting of the isolating system. Not only one, but two flexible connections were recommended in each piping circuit spaced at least 2 feet apart, preferably with one oriented in a horizontal run and one in a vertical run. This would effectively break up vibration wave patterns and greatly limit their transmission into adjacent structures.

After installation of the system shown in Figures 8 and 9, rigorous testing showed that it provided over 98% isolation. This meant that less than 2% of the significant reciprocating forces made their way into the surrounding floor and building structures. Therefore, it was still possible to successfully use the electron microscope on the second floor despite having a used two-stage reciprocating compressor operating directly beneath it, and only 21" away from the load bearing wall passing through these adjoining structures.

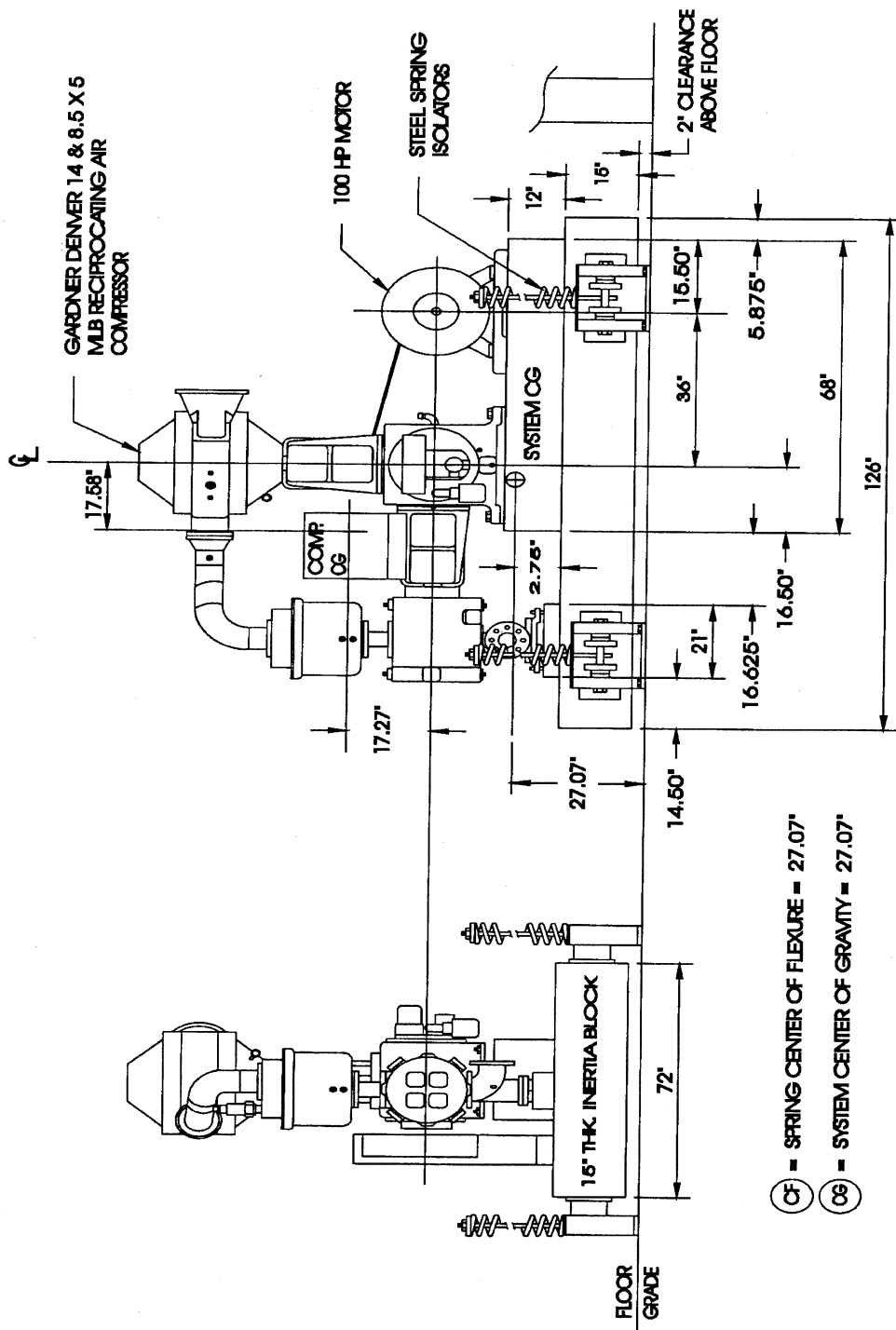


FIGURE 8
COMPRESSOR AND MOTOR ON PROPOSED VIBRATION ISOLATION SUPPORT SYSTEM
(Ref. 6)

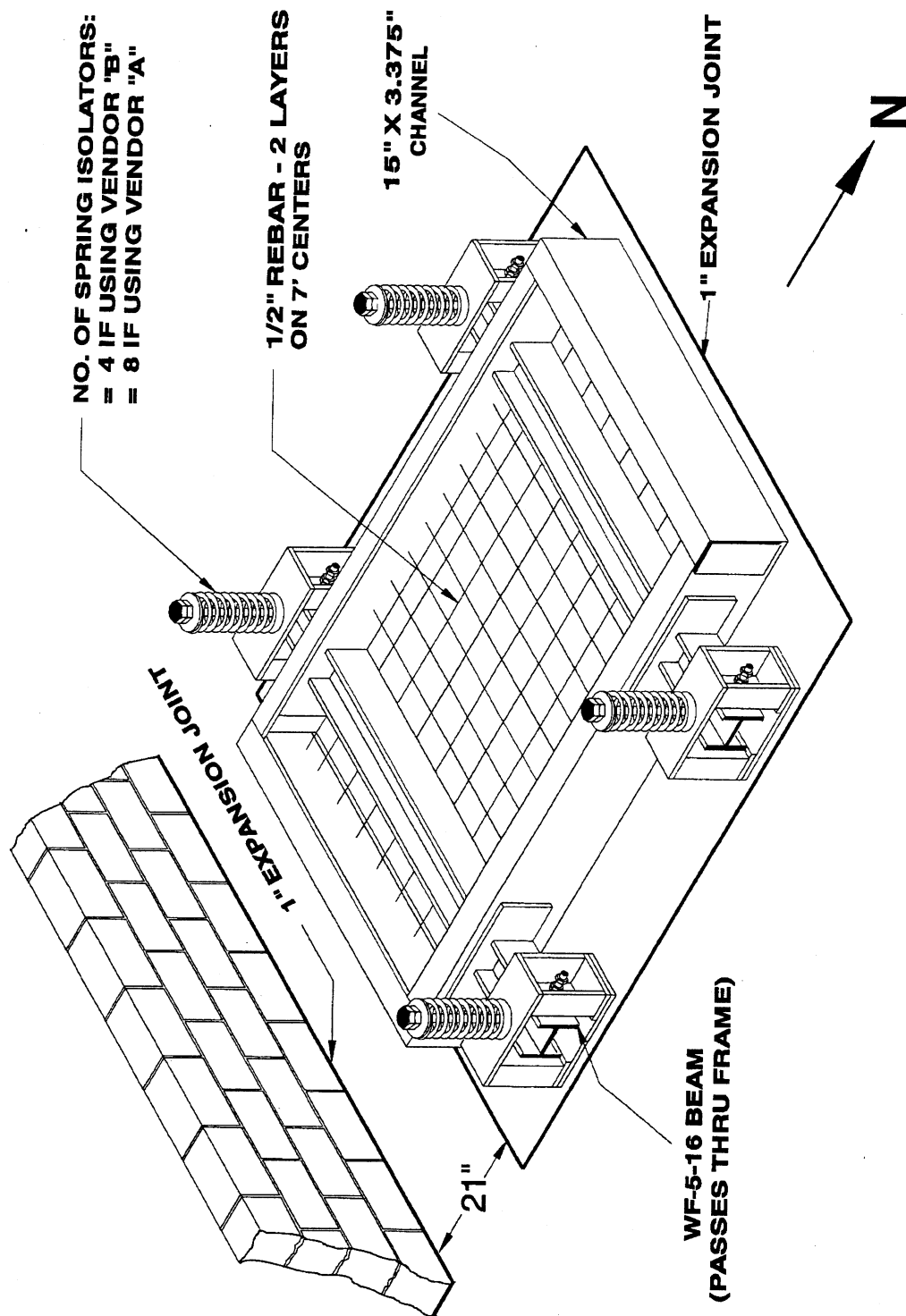
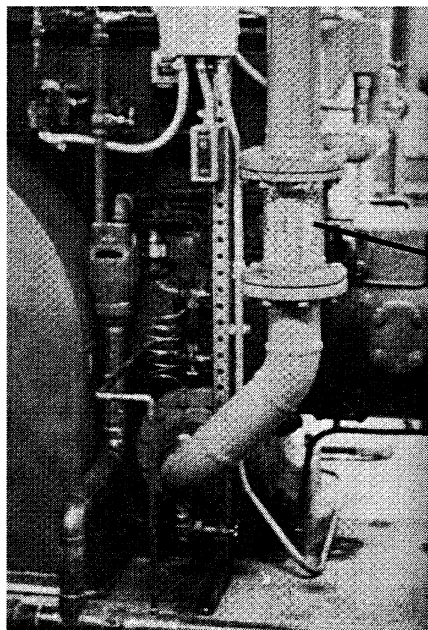


FIGURE 9
 PROPOSED VIBRATION ISOLATION FRAME

FIGURE 10
VIBRATION ISOLATION SYSTEM FOR A COMPRESSOR
AND ATTACHED PIPING (Ref. 7)



Flexible
Connection

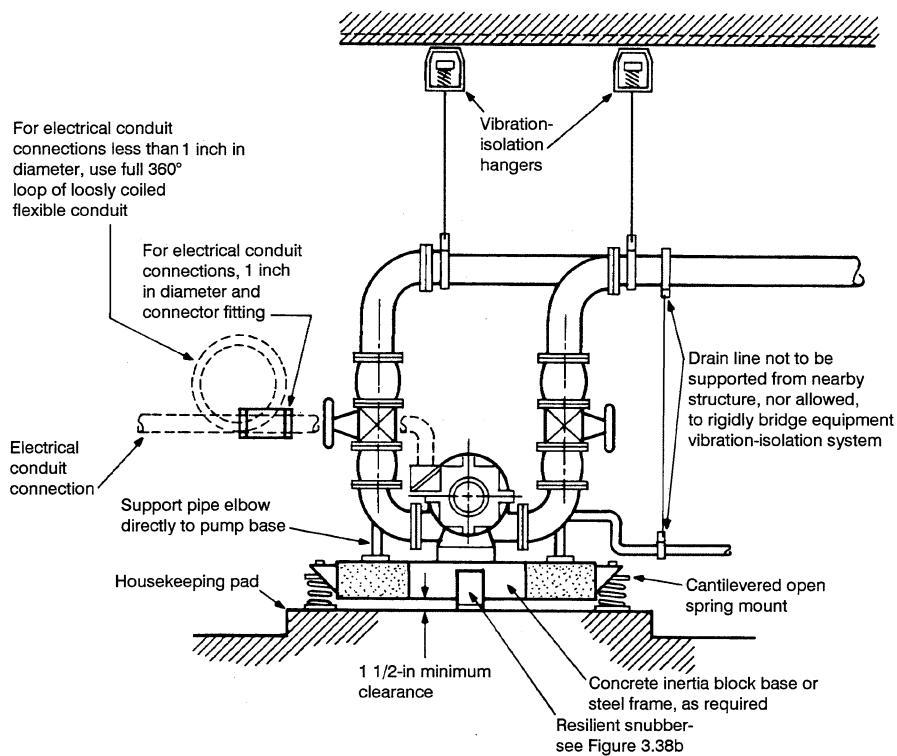


FIGURE 11
PROPERLY INSTALLED VIBRATION-ISOLATION PIPE HANGER
(Ref. 7)

Vibration-Isolation Hanger

The spring-in-series hanger with a neoprene element shown in Figure 3.105 is the proper type and is correctly installed. The isolator is clear of all nearby building construction and other utility systems. The unit may be easily inspected and adjusted; when installed correctly, it will function adequately.

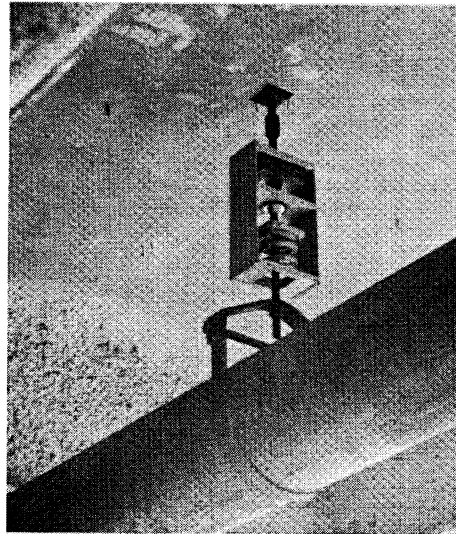


Fig. 3.105 Vibration-isolation hanger.

● **Rules of Thumb When Specifying Vibration Isolators -**

1. When specifying isolators for general rotating machinery, always ensure that the lowest significant forcing frequency is at least 3 times higher than the isolator natural frequency (in order to provide approximately 90% isolation). For more precision systems where more isolation is required, this ratio should be increased up to approximately 6 or 7.
2. Never specify an isolator having a natural frequency more than one-half the lowest significant frequency (don't forget the "belt RPM" in belt-driven systems; for example, in a 1780 RPM motor driving a 1000 RPM fan, the belt speed might be as low as 250 to 300 RPM which might prove to be a problem if five or six belts are required to drive the fan and problems develop with the belts). Therefore, it is always a good idea to take an analyzer into the field and capture a series of spectra, not only on the machine which is to be isolated, but also on and around other nearby machinery which might have even lower frequency vibration.
3. If isolating a direct coupled motor and pump, or a belt-driven fan, the analyst should consider installing a common base to support each of them (like that shown in Figure 8). Not doing so can introduce instability allowing each machine component to rock relative to one another.

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